

Vatsal Jha

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Research Interests

Decision-making under uncertainty and scarcity (bandits, RL); reliable and accountable ML systems (evaluation, robustness, calibration); agentic and federated systems for infrastructure and public-interest applications.

Education

Vellore Institute of Technology

2022–2026

B.Tech. Computer Science & Engineering, specialization in Internet of Things

Vellore, India

CGPA: 9.23/10. Top 1% nationally, NPTEL Reinforcement Learning (IIT Madras) – invited to interview for Instructor, Reinforcement Learning, IIT Madras BS Degree Program; Top 5% nationally, Responsible & Safe AI Systems (IIIT Hyderabad × IIT Madras).

Research & Engineering Experience

testAIng

Jan–Jul 2026

Machine Learning Intern

Remote

- Owned ML systems and research for multilingual audio-deepfake detection: ECAPA-TDNN, WavLM and mixture-of-experts models; internal multilingual benchmarks reached 0.94–0.96 AUC and Hindi fine-tuning reached 1.97% EER.
- Migrated model-heavy inference to Docker/Kubernetes with HPA, probes, rolling deployment and Prometheus/Grafana; built Playwright/Xvfb meeting analysis with WhisperX/Pyannote and a KEDA-scaled bot path. Placed 1st among 70+ interns in research presentation. [blog]

Samsung Research India

May–Jul 2025

Software Engineering Intern, Server Platform / MAGPIE

Bangalore, India

- Individually designed and built a multi-agent root-cause analysis prototype from literature review through evaluation/handoff, using Google ADK and MCP tools over Prometheus, Kubernetes, Jaeger, OpenSearch and Milvus.
- Implemented modality-specialist evidence collection, hypothesis validation/halting, cross-incident retrieval memory, context reduction and diagnosis-only safety boundaries; built an OpenTelemetry Demo + Chaos Mesh fault-injection testbed. Received a full-time Pre-Placement Offer (PPO).

Samsung R&D Institute Bangalore – PRISM

Oct 2024–May 2025

Team Lead / Co-primary Researcher

Hybrid

- Co-led four student researchers on HOMA, a multilingual SLM-orchestrated smart-home system that decomposes multi-device commands and coordinates sequential/concurrent execution across 11+ languages.
- Built synthetic multilingual evaluation and benchmarked Gemma/Qwen/Phi-family models under edge constraints. Samsung PRISM Excellent Worklet Award; work presented at ICCNT 2025. [code]

Vellore Institute of Technology – Dr. B. Ashok

Aug 2024–Present

Researcher

Vellore / Remote

- Research spanning federated in-vehicle recommendation, simulator-in-the-loop EV infrastructure planning, selection-aware carbon claims, and robustness/decision-making under heterogeneous or shifted data.
- Co-inventor on Indian Patent Application 202641089937 (Published) for privacy-preserving federated in-vehicle recommendation (BehaviorFL/BehaviorFL-MT) under heterogeneous sensors and partial task supervision; discovery-aware client sampling gives a $5.8\times$ increase in rare-task-client exposure at a fixed communication budget, evaluated across three public mobility datasets (VED, GeoLife, T-Drive).
- Leads HERO, a two-stage EV-charging-station placement system: an XGBoost city-level demand model feeds a simulator-in-the-loop multi-armed-bandit search that treats each complete station layout as a persistent arm; outperforms 13 facility-location, metaheuristic and bandit baselines by up to 39.9% composite reward in Ann Arbor SUMO microsimulation, with zero-shot cross-city demand transfer validated by rank correlation.
- Audited carbon-claim overstatement arising from post-selection bias in real supplier EPD data; a selection-aware reporting bound reduces overstatement from 51% to 11% without changing the recommended product.

NTNU × BITS Pilani

Mar 2024–Aug 2025

Remote Research Collaborator, with Dr. Sukalpa Chanda and Aniket Anand Gurav

Remote

- Built an end-to-end handwriting detection+recognition pipeline on IAM: shared ResNet-50 backbone, FOTS-style region detector, rotated ROI alignment, and an attention-augmented PHOSC word recognizer (custom 8-head scaled dot-product attention), jointly trained under constrained compute; selected internal development/test runs reached roughly 6–8% CER.

Council on Energy, Environment and Water (CEEW)

Jun–Aug 2024

Data Science Intern

Delhi, India

- Built NLP over 93K performance-criteria/task segments from 600+ occupations using BERTopic, transformer embeddings, UMAP/HDBSCAN and semantic matching; shipped a public job-transition prototype that later informed the public CEEW–ADB SMART platform and a 2026 CEEW coal-workforce report. [HF] [SMART] [report]
- Co-developed ML bias correction for Delhi WRF-CAMx forecasts: PM2.5 MAPE 43%→27%, RMSE 73→42 $\mu\text{g}/\text{m}^3$, correlation 0.70→0.78; coauthored IICAQM 2024 poster.
- Implemented a modular 3.1K-line Python/Streamlit solar/BESS techno-economic decision tool covering dispatch, tariffs, outages, sizing and multi-year financial modeling. [code]

- Designed a stochastic trip-segmentation pipeline and ML methodology for electric two-wheeler battery SoC prediction using real on-road CAN/BMS telemetry, under a Royal Academy of Engineering Distinguished International Associate grant (DIA-2022-150) with King Saud University; test $MSE \approx 0.041$ (Voting Regressor). Published in *Journal of Energy Storage*.

Peer-Reviewed Publications

- V. Jha, Y. Babu S. “CA-AMFA: Context-Aware Adaptive Multi-Factor Authentication for IoT-Based Household Security Systems.” *IEEE ICDSINC*, 2025. **Best Paper Award**. [IEEE]
- A. Jain, V. Jha, F. Alsaif, B. Ashok, I. Vairavasundaram, C. Kavitha. “Machine learning framework using on-road realtime data for battery SoC level prediction in electric two-wheelers.” *Journal of Energy Storage*, 97, 112884, 2024. DOI: 10.1016/j.est.2024.112884. [ScienceDirect]

Accepted / Presented; Proceedings Pending

- A. Jain, V. Jha, et al. “HOMA: Home Orchestration using Multi-Agent systems.” ICCCNT 2025. Presented; proceedings appearance pending/finalizing. [code]
- S. Murugaiyan, A. Venkatesa, V. Jha, A. Jain. “Nonlinear Model Predictive Control for Dynamic Obstacle Avoidance in RoboCup SSL.” HINT’25. Accepted/presented; Springer LNNS proceedings pending. [code]

Posters / Extended Research Outputs

- S. Tiwari, V. Jha, M. Rafiuddin, S. K. Guttikunda. “Bias Correction of Forecasts from a Chemical Transport Model using Machine Learning for Delhi.” ICAQM 2024, IIT Madras. Poster.

Under Review / Submitted Research

- V. Jha. “The Priority Stability Index: A Pre-Deployment Audit Instrument for Seed Instability in Predictive Policing.” AAAI 2027 AI for Social Impact Track. Under review. Single author.
- V. Jha. “Diagnosing Identifiability Failure in Reporting-Adjusted Crime Models: A Template Applied to India’s Recorded Rape Data.” AAAI 2027 AI for Social Impact Track. Under review. Single author.
- V. Jha, A. Srivastava, B. Ashok. “Ranking Is Not Valuation: Selection-Aware Carbon Claims under Supplier Shift.” NeurIPS 2026 Tackling Climate Change with Machine Learning Workshop. Submitted.

BehaviorFL-MT: Federated Multi-Task Learning under Unequal Vehicle Sensors and Partial Supervision. NeurIPS 2026 NewInML Workshop. Submitted/under review.

SOPRA: Context, Not Scaffolding: A Simplifying Finding for LLM-Based Java Test-Case Repair. SANER 2027. Under review.

HERO: A Hybrid Machine Learning and Multi-Armed Bandit Framework for Electric Vehicle Charging Station Placement via High-Fidelity Traffic Simulation. Engineering Applications of Artificial Intelligence. Under review.

Ongoing Research

CA-AMFA journal extension. Single-author working paper, “CA-AMFA: Label-Free Online Risk-Weight Adaptation for Multi-Factor Authentication in Residential IoT”; decomposes adaptive gains from static recalibration, adds public CASAS-stream validation and adversarial stress tests; in preparation for JISA submission.

ResourceArena / Semantically Strategyproof Allocation. Building a reusable environment for strategic, language-mediated scarce-resource allocation, separating representation sensitivity from strategic controllability; active falsification with AAMAS 2027 as target.

Patents & Intellectual Property

B. Ashok, Vatsal Jha. “System and Method for Privacy-Preserving In-Vehicle Recommendation using Federated Learning and Cross-Domain Feature Alignment.” Indian Application No. **202641089937**, filed 24 Jul 2026; VIT IPR IPR0004652P.

Selected Software & Engineering Projects

Eldermere. A multiplayer systems project: built/deployed a Go + SvelteKit + PostgreSQL browser MUD with WebSocket room/presence state, persistent sessions/player state, data-driven JSON content packs, validation/tests, Docker Compose and procedural visual/ambient-audio generation. [code] [live] [docs]

ChatterBot. A scalable LLM-serving prototype implemented with FastAPI + Ray Serve/vLLM + Redis + RabbitMQ + PostgreSQL, with synchronous, queued asynchronous and SSE streaming modes, context caching, AWQ path, retries/circuit breakers and Locust load tests; architecture designed for 10K-concurrency horizontal scaling. [code]

CtrlAltHeal / Vita Sana. Freelance platform engineering: sole engineer of two parallel backend stacks (Node/Express/Prisma/PostgreSQL diet platform; a separate C# ASP.NET/EF Core management platform) behind 5 client/handover surfaces; built DOCX/XLSX/CSV pipelines converting legacy nutrition/PCOS clinical documents into structured data via Jaccard/token-set fuzzy matching; produced full architecture, API, security and operational-runbook handover documentation.

Narad (IISF–ISRO/NRSC Bhuvan). National-winning 22-language voice/query system; owned personalized recommendation and model fine-tuning/detoxification; system used AI4Bharat, FAISS, FastAPI and quantized models. [code] [challenge]

Maieutic Research. MIT-licensed Claude Code skill for Socratic human-AI research collaboration, with CORE/COLLAB/GRUNT decision zoning, prediction logs and bounded autonomy. [code]

Honors & Awards

- Best Paper Award, IEEE ICDSINC 2025, for CA-AMFA (first author).
- Samsung PRISM Excellent Worklet Award, for HOMA.
- National Winner, IISF 2023 Students Innovation Festival Space Hackathon (ISRO/NRSC Bhuvan challenge; 4,000+ registered teams); National Winner, Samsung PRISM hackathon; Top-3 finalist, Karnataka State Police Datathon (380 teams).
- 1st Prize, testAIing intern research presentation competition, among 70+ interns.
- Raman Research Award, VIT, for EV SoC research.

Leadership & Service

Co-Founder & Co-Team Lead, Team Prometheus, VIT RoboCup Small-Size League: founded VIT's first SSL team from zero; built and mentored a 30+ member robotics research/engineering group and secured major in-kind engineering-software sponsorships (SolidWorks, ANSYS, Altium).

Youth Red Cross, VIT: Vice Chairperson (2024) and Chairperson / student leadership (2025); facilitated 750+ blood donations during Chairperson year and coordinated community/service activities.

Technical Skills

ML/Research: PyTorch, TensorFlow, Hugging Face, scikit-learn, Flower, RL/bandits, federated learning, LLM/agent systems, RAG, statistical evaluation, experiment design.

Systems: Docker, Kubernetes, KEDA, FastAPI, Ray Serve/vLLM, MCP/Google ADK, PostgreSQL, Redis, Milvus, RabbitMQ, Prometheus/Grafana, OpenSearch, Jaeger, SUMO.

Languages: Python, Java, C++, Go, SQL, JavaScript/TypeScript, C#.